

FDSS[®]-GX

Kinetic Plate Imager
C15711-02



HAMAMATSU
PHOTON IS OUR BUSINESS

Kinetic Plate Imager

for reliable and stable high-throughput screening (HTS)

Since the 1990s, Hamamatsu Photonics has designed 9 models of kinetic plate imagers and has been providing solutions for kinetic, cell-based assays mainly for GPCR/ion channel research and screening in the drug discovery market.

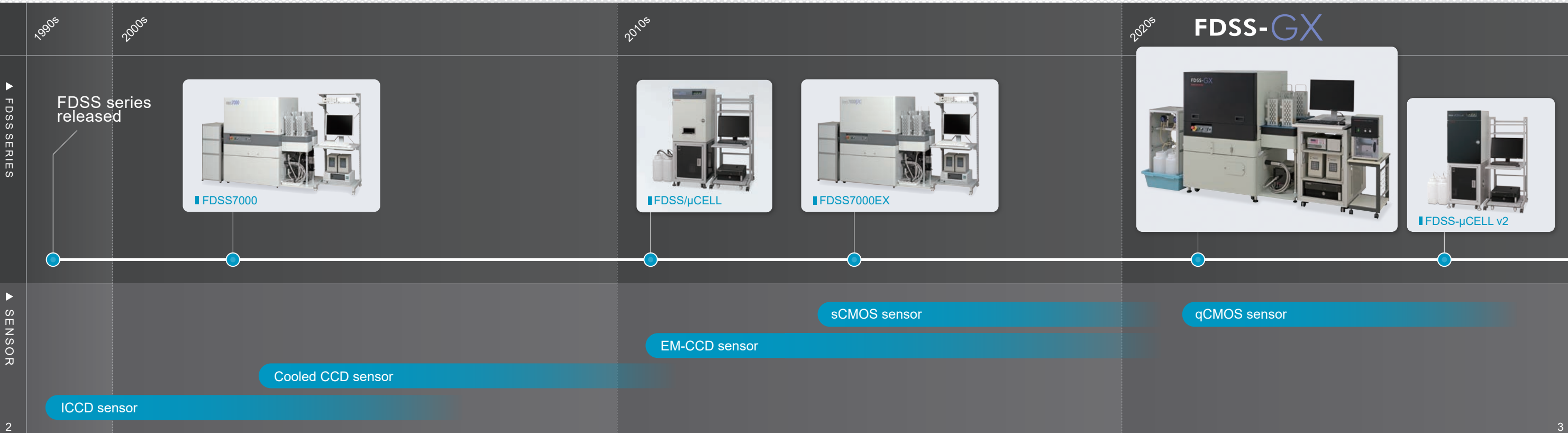
In 2022, Hamamatsu Photonics has released the FDSS®-GX, the highest-performance kinetic plate imager in the FDSS series, maximizing the technology and experience gained from the field of drug discovery.

FDSS-GX



POINT	High-precision epifluorescence optical system
01	1536 ch manifold pipettor head with independent metal piston cylinders
02	High-sensitivity qCMOS® sensor for fluorescence and luminescence
03	Semi-automation with stackers

Enabling whole microplate imaging and simultaneous injections to all wells >>> Well-optimized fluorescence and luminescence system for HTS



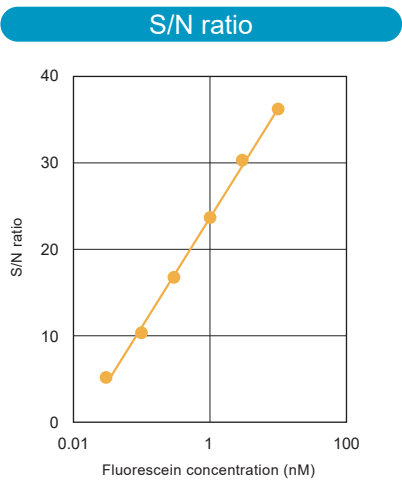
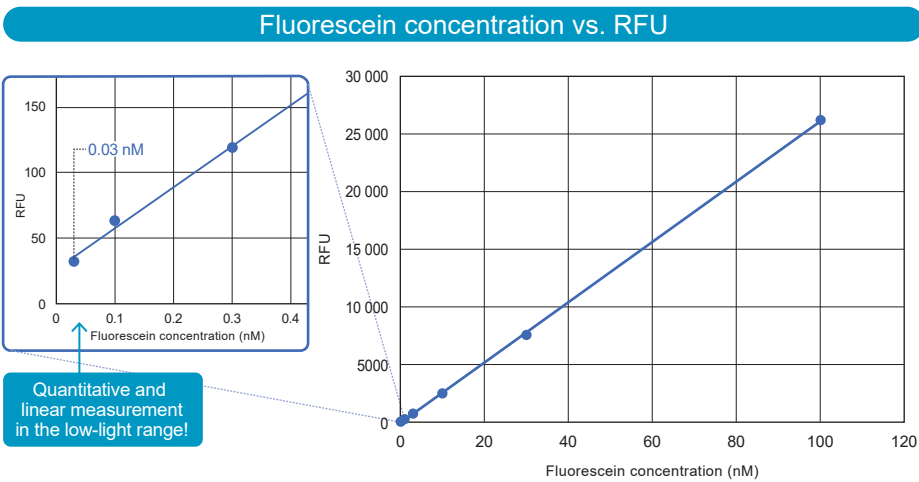
Quantitative and wide assay window

The high-sensitivity qCMOS sensor comes as a standard in the FDSS-GX optical system. In fluorescence and luminescence measurements, high quantitative performance has been achieved for low-light imaging, and the assay window has been expanded due to the sensor's wide dynamic range, enabling measurement with a wide linearity range.

>> Detection sensitivity in fluorescence

In fluorescence measurements, the FDSS-GX was able to detect 0.03 nM Fluorescein* (exposure time: 50 ms). The graph below shows the fluorescence count and S/N ratio for titrated concentrations of Fluorescein, and the image on the right shows the 384 microplate.

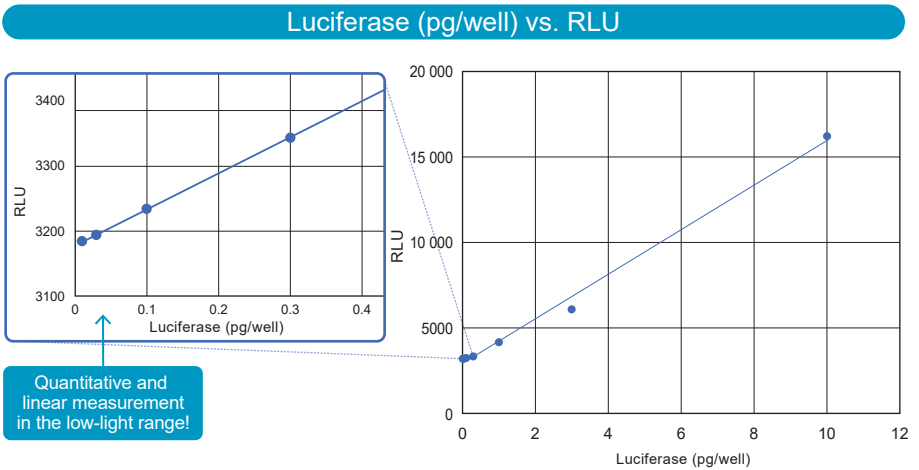
* This is the actual measurement value.



>> Detections sensitivity in luminescence

In luminescence measurements, the FDSS-GX was able to detect 3.9 fg Luciferase* (exposure time: 5 seconds) according to the calculations. The graph below shows the luminescence count for titrated concentrations of Luciferase, and the image on the right shows the 384 microplate.

* This is the theoretical value calculated from the measurement data below.

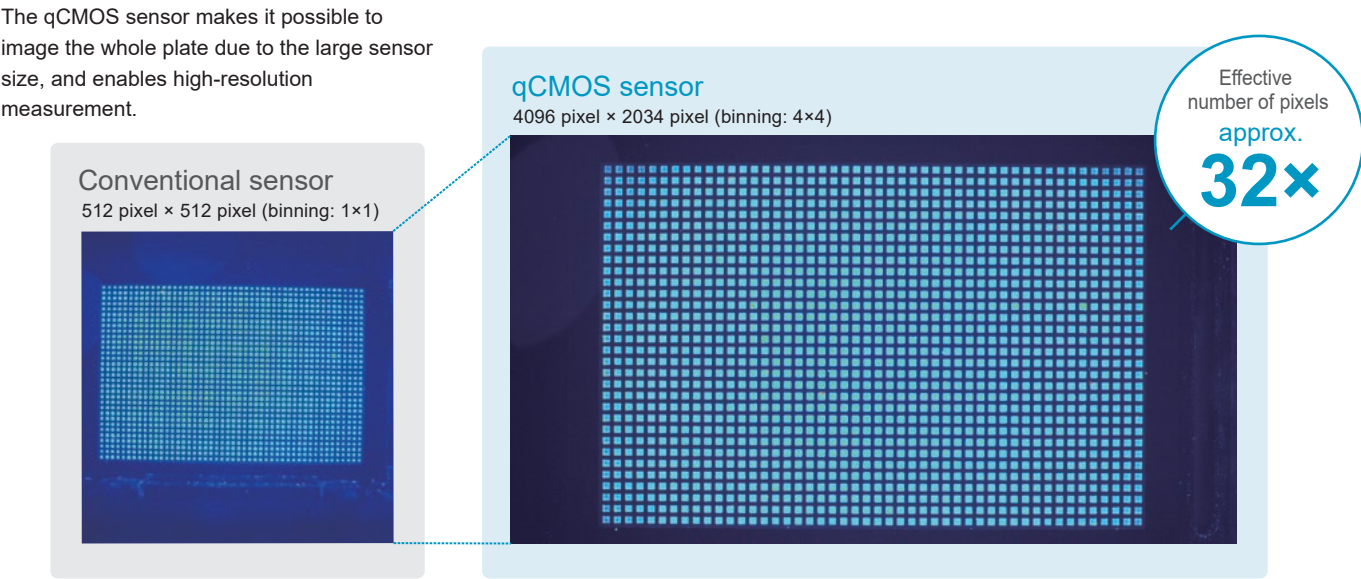


High-resolution whole plate imaging

Kinetic plate imagers that use the qCMOS sensor have many advantages due to the large sensor size and low noise. The qCMOS sensor enables measurement with high sensitivity and high resolution.

>> Comparison of whole plate imaging

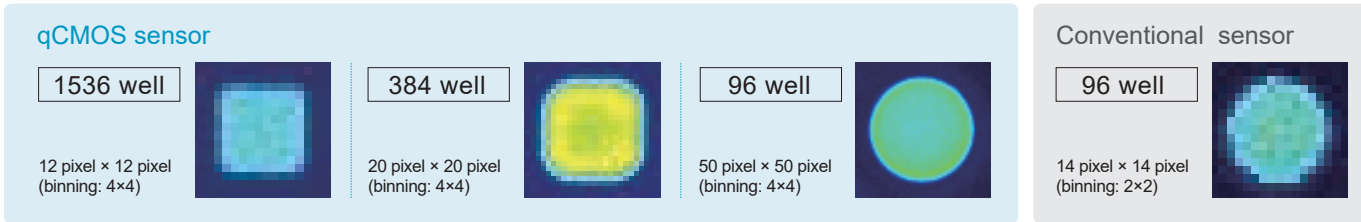
(1536 microplate, Sample: B-Beads)



>> Comparison of well image

(Sample: B-Beads)

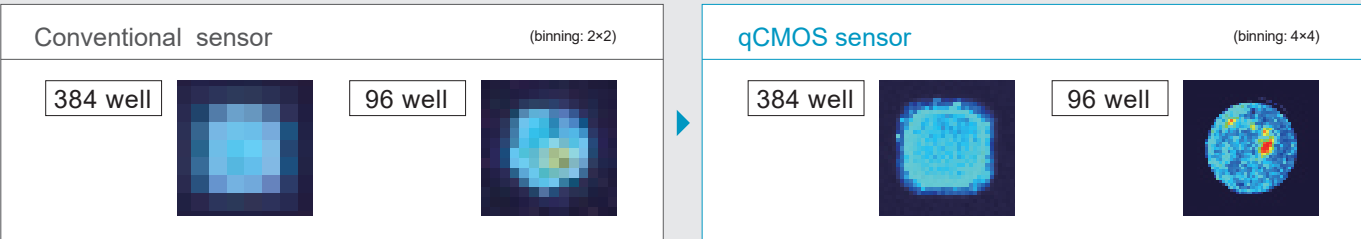
For kinetic plate imagers that measure average intensity in wells, it is essential to set the ROI (Region of Interest) precisely for quantitative measurement. Compared to the conventional sensor, the number of pixels per well has increased and the ROI is set more accurately, enabling quantitative measurement.



Observation in wells

(384 well / sample: primary rat cortex neuron cells / Fluo-4, 96 well / sample: CHO cell)

The high-resolution and low-noise measurements allow observation inside the wells.



1536 ch manifold pipettor head with independent metal piston cylinders

FDSS-GX

Simultaneous injection to all wells in 1536 plates

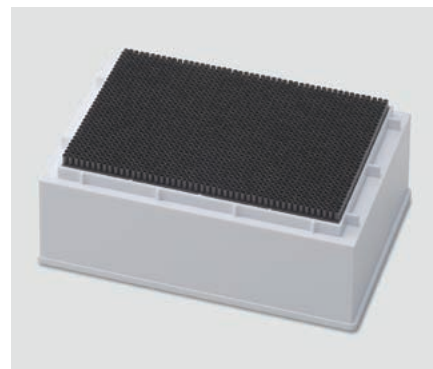
Dispensing accuracy and repeatability are important factors in performing assays.

The FDSS-GX achieves highly accurate and repeatable micro-dispensing by adopting a 1536 ch dispensing unit capable of dispensing variable volumes. This dispensing unit has independent metal piston cylinders and dispensing tips dedicated to the FDSS-GX.

* Dispensing units in 96 ch and 384 ch are also available.



Dispensing unit (1536 tip type) A15623-28



1536 black tip (10 racks) for FDSS-GX A8687-82

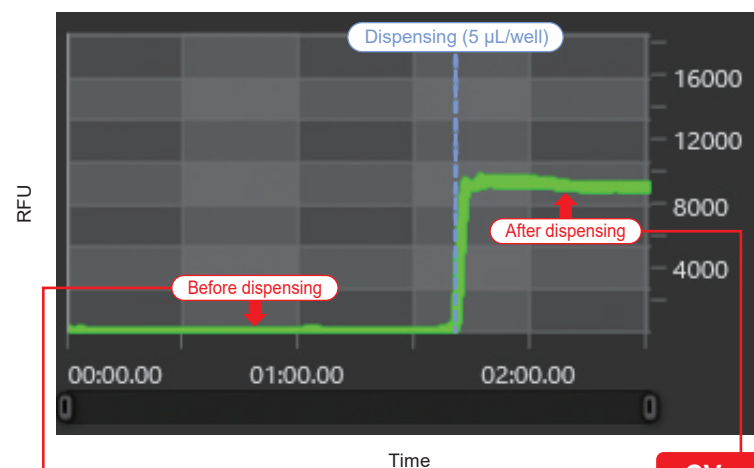
Dispensing performance

The figure below shows the fluorescence intensity change in time and plate image when B-beads were dispensed into a 1536 microplate (5 μ L/well), showing a CV of 1.89 %* after dispensing.

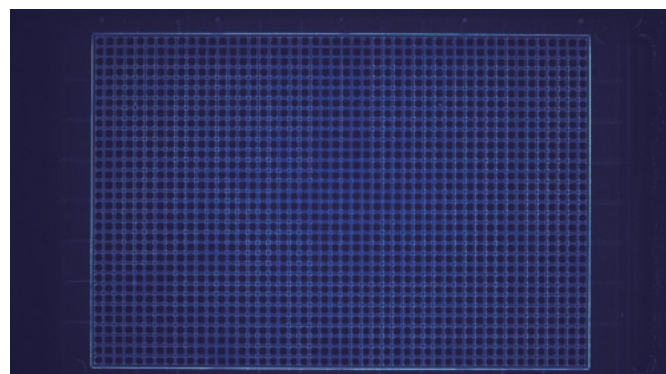
The 1536 ch dispensing unit with independent metal piston cylinders and dispensing tips dedicated to FDSS provide the best dispensing performance.

* This is the actual measurement value when dispensing B-beads (5 μ L/well) with overhauled or calibrated 1536 ch dispensing unit and dispensing tips dedicated to FDSS-GX. Please refer to page 12 for detailed specifications.

Fluorescence intensity change in time



Before dispensing (1536 microplate)



After dispensing (1536 microplate)



Advanced washing stage

In constructing an assay system for HTS, washing dispensed tips enables the reuse of the tips, thereby reducing costs and improving efficiency of an assay.

The FDSS-GX is compatible with automatic washing up to 3 solvents and achieves washing with no carry over.

* The dispensing accuracy is not guaranteed for reuse after the tips are unloaded.

Tip washer

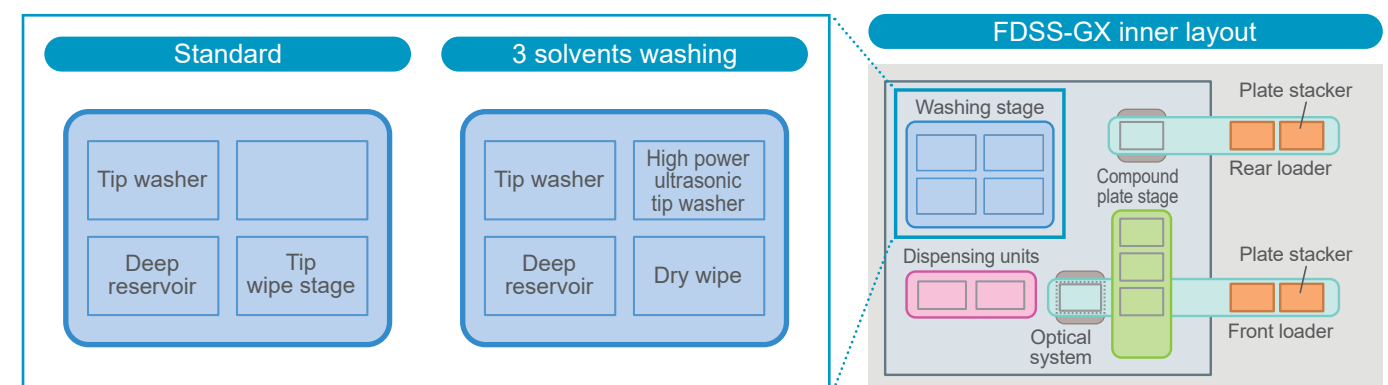
- Up to 3 tip washers available
- Overflow washing with chimney plates
- High power ultrasonic tip washer* available

Tip wipe stage

- Tip wipe stage is equipped as standard to blot out water droplets on tips after washing
- Dry wipe* to always keep blotting capability by suction pump is available

* Optional component

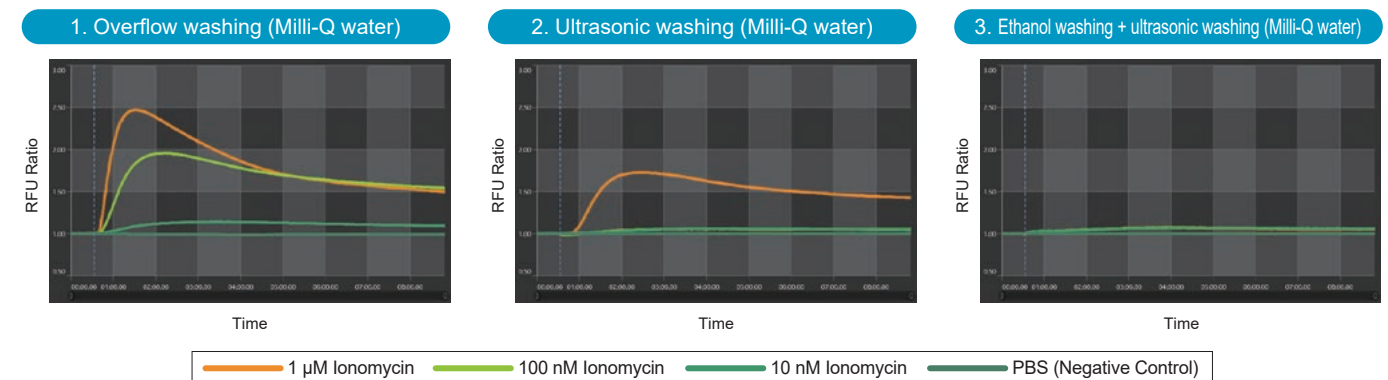
Example of washing unit configuration



Washing effect of high power ultrasonic tip washer

Tip washing with the high power ultrasonic tip washer prevents carryover of compounds that could not be washed out by overflow washing, reducing tip costs and improving assay efficiency.

The figure below shows data regarding the washing effect of overflow washing and ultrasonic washing using Milli-Q water. After attaching each concentration of Ionomycin to dispensing tips, tips were washed under the following conditions and the presence of carryover was checked.



* Each washing involves 10 cycles of aspiration and dispensing operation.
* Use 50 % ethanol solution

Normal overflow washing does not fully wash concentrations of 1 μ M and 100 nM Ionomycin. When ultrasonic washing is performed, 100 nM Ionomycin is completely washed out and no carryover is observed. Furthermore, it can be seen that even very high concentrations of 1 μ M Ionomycin are completely washed when washed with an ethanol solution prior to ultrasonic washing.

Measurement flow for semi-automation

FDSS-GX

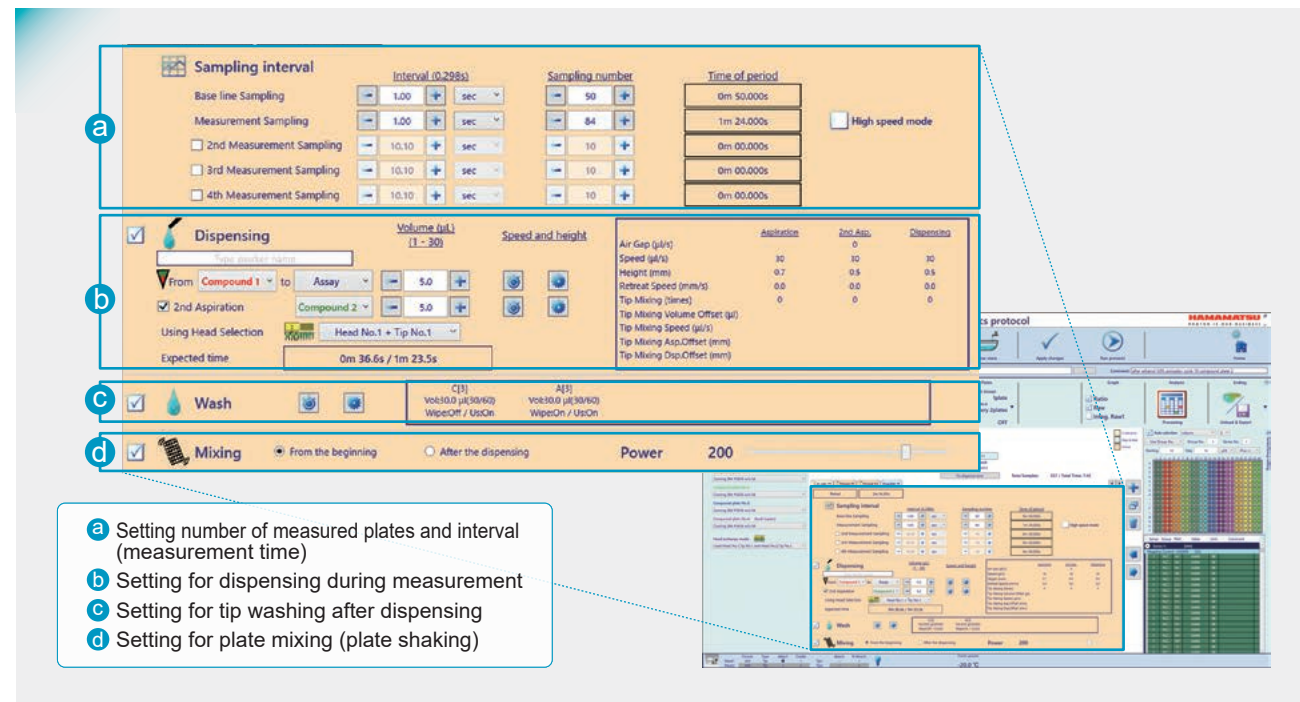
Load the FDSS dedicated dispensing tips to dispensing unit by using automatic tip loader and set the tip-loaded dispensing unit into the FDSS-GX main unit.



Tip loading

Head setting

Plate setting



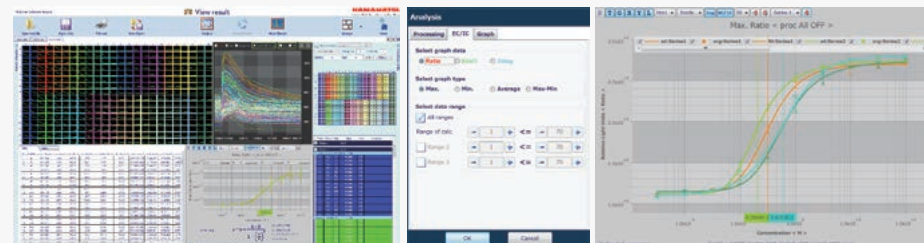
- a Setting number of measured plates and interval (measurement time)
- b Setting for dispensing during measurement
- c Setting for tip washing after dispensing
- d Setting for plate mixing (plate shaking)

Protocol setting

Data acquisition

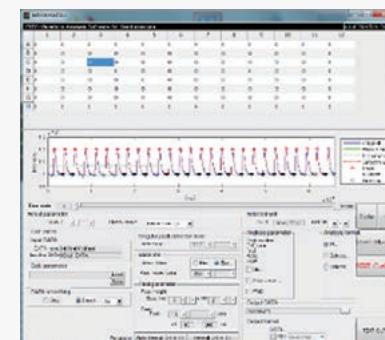
Data analysis

Various data processing and analysis of measurement results are possible.



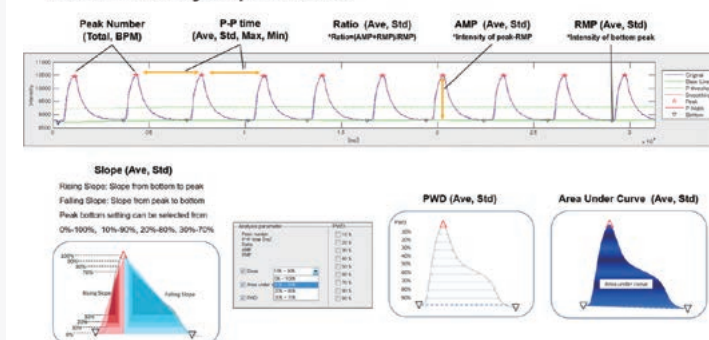
- Spatial correction between wells (spatial uniformity)
- Negative control correction
- Positive control correction
- Baseline subtraction correction (subtract bias)
- IC/EC graph calculation from multiple series (4 or 5 parameters may be selected)
- IC/EC graph calculation using Max, Min, Average and Max-Min in up to 3 time ranges in the same series
- Slope calculation to maximum range of 8
- Max, Min, Max-Min and Ratio calculation to maximum range of 8

Analysis of calcium transient waveform of iPS cardiomyocyte



- Waveform peak number (Peak number: Total, BPM)
- Peak-to-peak time (p-p time: Ave, Std, Max, Min)
- Peak luminescence value/bottom luminance value ratio (Ratio: Ave, Std)
- Peak amplitude (Peak luminance value) (Amplitude: Ave, Std)
- Bottom luminance value (RMP: Ave, Std)
- Rise and fall slope (Rising/Falling slope: Ave, Std)
- Peak pulse width 10 % to 90 % (PWD10, 20, 30, 40, 50, 60, 70, 80, 90)
- Peak total area (Area under curve: Ave, Std)

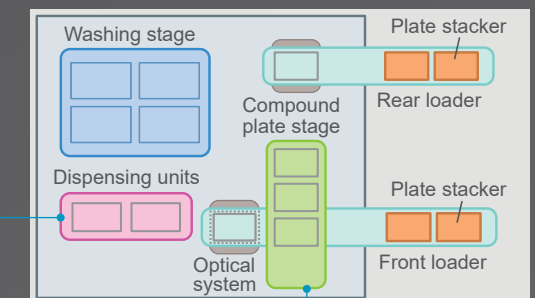
Waveform analysis parameters



A wide variety of assay designs available

Two dispensing units capability

- Two dispensing units can be used according to the characteristics of compounds / reagents
- Compatible with 1536 / 384 / 96 tip type
- Dispensing units can be selected depending on assay design



Compound plate stage

- 3 fixed compound plate stages available
- Compound plate loading using rear loader
- Automatic reagent feeder* can be equipped on the fixed compound plate stage

*Optional component

Highly flexible plate transfer

Plate stacker unit*

- Semi-automation can be achieved by automatic transfer of assay plate and compound plate
- Large capacity of cassette for up to 20 microplates
- Barcode reader* for reading the barcode attached to the assay / reagent plate is selectable
- Lid opener function available



*Optional component

Robot connection

- Automation accessories enable connection to robots from various vendors

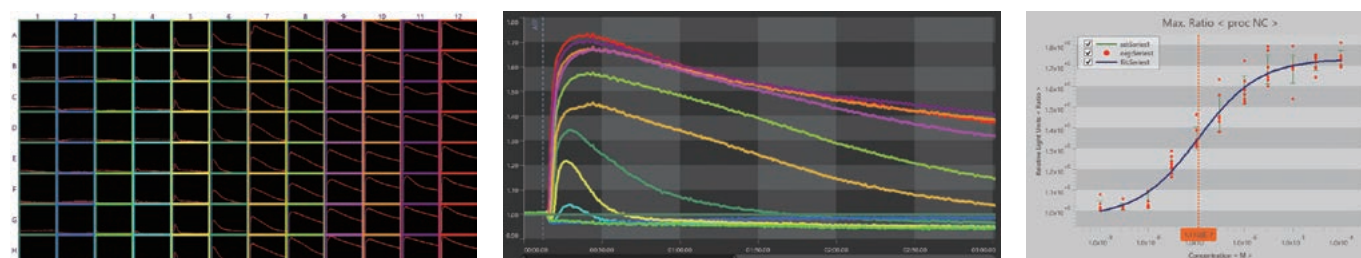


Fluorescence applications

Ca²⁺ assays

Fluo-8, Calbryte -520, -590, -630 AM, Fura-2

Evaluation of ATP dose response using Fluo-8 AM-stained CHO cells



- Cell: CHO cell
- Dye: Fluo-8 AM (AAT Bioquest)
- Compound: ATP final 100 μ M –1 nM

Membrane potential assays

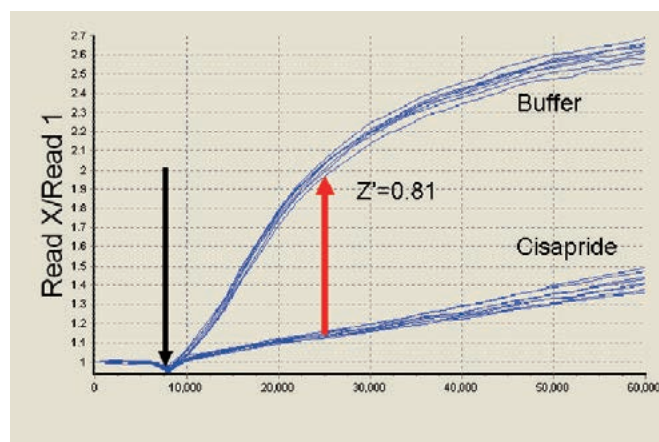
FluoVolt, SQMP, FMP

Other ion channel assays

K⁺: FluxOR*, Na⁺: ANG2, SodiumGreen, Cl⁻: YFP, diH-MEQ

*Potassium ion channel screening using TI⁺ is available.

K⁺ assay in CHO cells using FluxOR



- Cell: CHO cell
- Dye: Flux-OR Potassium

iPSC-derived cells

Ca²⁺ transient and membrane potential measurements using iPS cell-derived cardiomyocytes

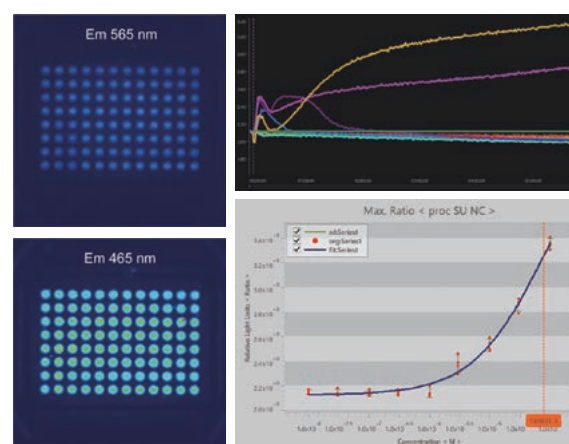


- Cell: iCell Cardiomyocytes² (CDI)
- Dye: Cal-520AM
- Ca²⁺ transient after addition of various compounds

FRET assays

CFP/YFP, voltage sensor probe, fluorescence probe, fluorescence protein

Evaluation of Nav 1.5-CHO cells using FRET-type voltage sensitive dye (VSP)



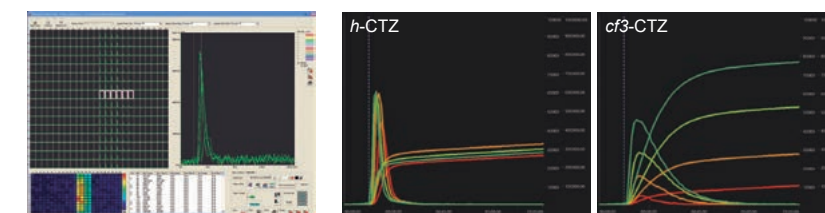
- Cell: Nav1.5-CHO cells (Ion Chat Research Corporation)
- Dye: Donor: CC2-DMPE (Invitrogen) final 5 μ M
- Acceptor: DiSBAC4(3) (Invitrogen) final 10 μ M
- Compound: Veratridine (Sigma) final 100 μ M –10 nM

Luminescence applications

Ca²⁺ assays

Aequorin

Intracellular Ca²⁺ assay by luminescence using an aequorin-expressing cell line



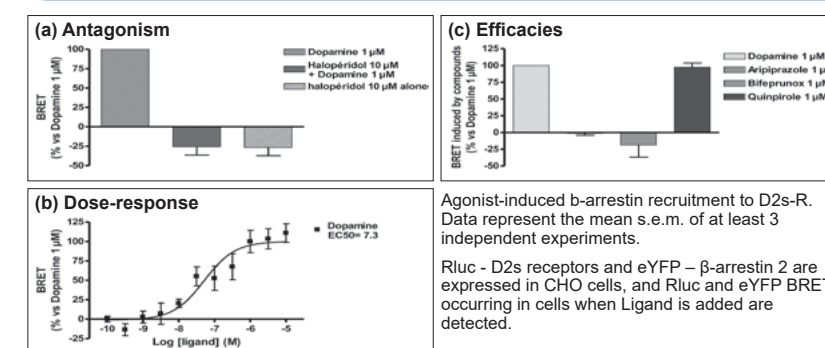
- Cell: Aeg-CHO (8000 cells/well)
- Substrate: coelenterazine
- Ligand: ATP (500 nM, 100 nM, 20 nM)
- Cell: CHO-K1 stably expressing apoaequorin with a mitochondrial targeting signal
- Substrate: h-coelenterazine (h-CTZ), cf3-coelenterazine (cf3-CTZ)
- Compound: acetylcholine final 30 nM–1 μ M

S. Inouye, R. Iimori, Y. Sahara, S. Hisada, T. Hosoya, Application of new semisynthetic aequorins with long half-decay time of luminescence to G-protein-coupled receptor assay, *Analytical biochemistry*. 407. 2, 247-252 (2010).

BRET assays

BRET1, BRET2, NanoBRET

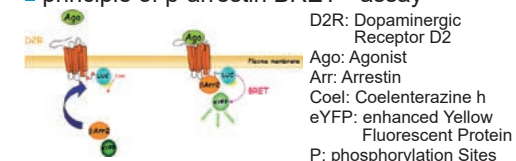
BRET assay using CHO cells



Agonist-induced b-arrestin recruitment to D2s-R. Data represent the mean s.e.m. of at least 3 independent experiments.

Gluc - D2s receptors and eYFP - β -arrestin 2 are expressed in CHO cells, and Rluc and eYFP BRET occurring in cells when Ligand is added are detected.

principle of β -arrestin BRET¹- assay

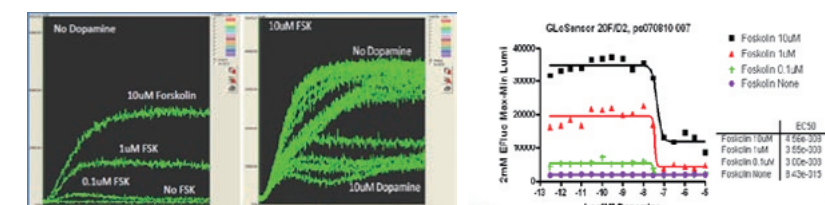


Data courtesy : Frederic Finana
Biologie Cellulaire et Moléculaire, Centre de Recherche Pierre Fabre Finana F, De Vries L, Raully-Lestienne I et al. 10th European Functional Drug Screening Symposium Poster No.8 (2014)

cAMP assays

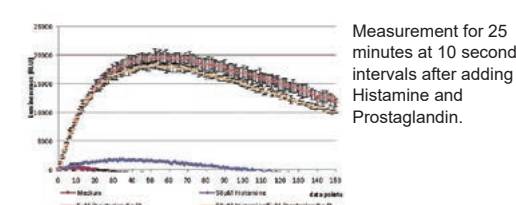
GloSensor

cAMP evaluation using GloSensor



- Cell: CHO cell
- Kit: GloSensor

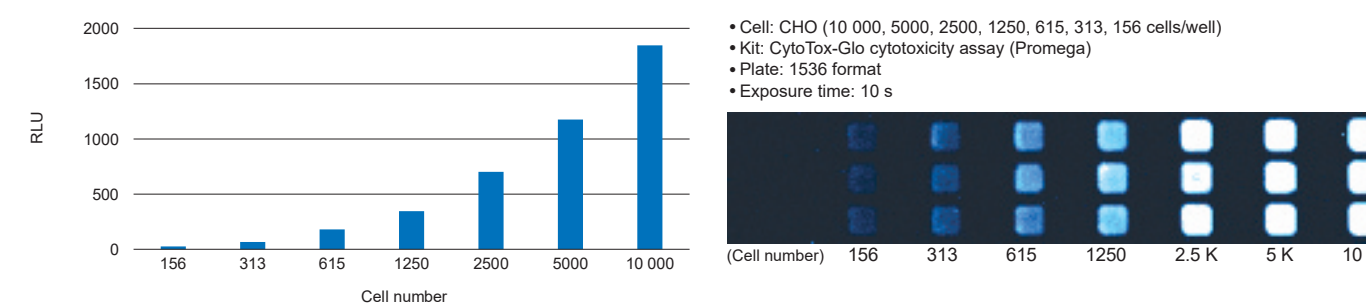
Analysis of time course of cAMP using HUVEC expressing GloSensor (Promega)



- Cell: HUVEC
- Kit: GloSensor

Cell number evaluation using CytoTox-Glo cytotoxicity assay

Cell number evaluation using CytoTox-Glo cytotoxicity Assay



- Cell: CHO (10 000, 5000, 2500, 1250, 615, 313, 156 cells/well)
- Kit: CytoTox-Glo cytotoxicity assay (Promega)
- Plate: 1536 format
- Exposure time: 10 s

FDSS-GX standard configuration

FDSS-GX Kinetic Plate Imager C15711-02

- **Fluorescence luminescence sensor unit (1 lamp)**
 - Xe lamp, Automatic EX filter changer (5 positions for filters)
 - Automatic EM filter/DM slot changer (2 positions for DM, 4 positions for filters)
 - Excitation filters: 472 nm (Fluo-4), 531 nm (FMP)
 - Emission filters: 520 nm to 560 nm (Fluo-4), 593 nm (FMP)
 - Dichroic mirrors: B (Fluo-4) / FMP
 - Shortpass filters: SPF 495 (Fluo-4), SPF 550 (FMP)
- **Detector**
 - Hamamatsu qCMOS sensor dedicated to FDSS
 - Circulating water cooler
 - Fluorescence/luminescence detection and measurement functions
 - Sampling rate: 10 Hz (10 data points per second) minimum
 - Sampling interval: 0.1 s minimum
- **Plate loading line**
 - Front loader and rear loader for loading of assay plate and compound plate
 - Assay plate: Front loader ×1
 - Compound plate: Compound plate stage (manual exchange) ×3 Rear loader ×1
- **Washing stage**
 - Tip washer (including bath, tubes, pump, tanks)
 - Deep reservoir
 - Tip wipe stage
 - Chimney plates (96 / 384 tip type)
- **Plate shaking function (front loader)**
 - Plate shaking level step: 256 (software control)
 - Revolutions per minute: 0 to 3000 rpm
- **Heater unit**
 - Keeps the main unit chamber in 35 °C to 37 °C range
 - Assay plate and compound plates are homogeneously at that temperature
- **FDSS software**
 - FDSS control software operates in data analysis unit
 - Compatible with Windows 64-bit OS



Data analysis unit C7903-13

64-bit computer, Windows® operating system for FDSS-GX

Automatic tip loader A15623-07

Automatically loading/unloading tips on to dispensing unit (1536, 384, 96 tip type).



Dispensing units

Dispensing unit (1536 tip type) A15623-28

Dispenser head for dispensing reagents simultaneously into a 1536-well microplate.
Dispensing volume: 1 µL to 5 µL, Dispensing accuracy: within CV 10 % (when dispensing 5 µL)
* including chimney plate (1536 tip type)



Dispensing unit (384 tip type) A10118-26

Dispenser head for dispensing reagents simultaneously into a 384-well microplate.
Dispensing volume: 1 µL to 30 µL, Dispensing accuracy: within CV 5 % (when dispensing 5 µL)



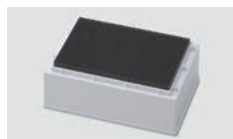
Dispensing unit (96 tip type) A10118-24

Dispenser head for dispensing reagents simultaneously into a 96-well microplate.
Dispensing volume: 10 µL to 200 µL, Dispensing accuracy: within CV 3 % (when dispensing 10 µL)

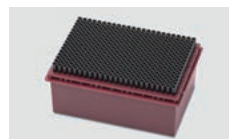


Consumables (Dispensing tips)

1536 black tip (10 racks) for FDSS-GX A8687-82



384 black tip (10 racks) for FDSS7000/µCELL/-GX A8687-62C



96 black tip (10 racks) for FDSS7000/µCELL/-GX A8687-32A



Options

Plate stacker set with lid opener A15623-05

Set of plate stackers for front/rear loader with lid opener. Microplates in cassette are automatically loaded in the FDSS-GX and unloaded after measurement. Storage capacity of 20 plates.



High power ultrasonic tip washer A15623-49

Unit for washing tips attached to the dispensing unit with high power ultrasonic. Includes bath, tubes, pump, washing liquid tank, waste liquid tank, ultrasonic controller.



Automatic reagent feeder A15623-55

Reagents are automatically supplied into the bath inside the FDSS-GX. Includes bath, pump, tubes, volume detection sensor, reagent tank rack.



* Reagent tank and stirrer are not included.

Deep reservoir A10118-61

Reservoir to be used with reagent or washing solution.



Filter set (CFP/YFP-FRET) A10343-21B

Excitation filter: 438 nm
Emission filter: 483 nm, 542 nm
Shortpass filter: SPF 450
Dichroic mirror: CFP/YFP-FRET



Additional tip washer A15623-48

Unit for washing tips attached to the dispensing unit. Includes bath, tubes, pump, washing liquid tank, waste liquid tank.



Barcode reader set for front and rear loaders A15623-50

Option for reading the barcode attached to the assay/reagent plate. Set for front/rear loader line.



Dry wipe A15623-56

Wipe stage to always keep blotting capability by suction pump is available.

* Pump is not included.



Filter set (VSP) A10343-01C

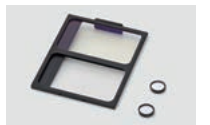
Excitation filter: 387 nm
Emission filter: 466 nm, 560 nm
Shortpass filter: SPF 400
Dichroic mirror: VSP



Filter set (Fura-2) A10343-61

Excitation filter: 340 nm, 387 nm
Dichroic mirror: Fura-2
ND filter 0.3

* Use the emission filter and the shortpass filter for Fluo-4 included in C15711-02.
Emission filter: 520 nm to 560 nm
Shortpass filter: SPF 495 (Fluo-4)



* Other filter sets or single filters are also available.
Please contact your Hamamatsu representative for further information.

Software options

FDSS software Additional offline software license U8524-03A

Used to display, analyze and output data on devices other than FDSS-GX. Windows 64-bit OS compatible.

FDSS software option Waveform analysis software for cardiomyocyte U8524-12

Software protection key for multi-well analysis of waveform obtained from cardiomyocytes.

FDSS software option Export TIFF image option U8524-14

Add function to save TIFF (16-bit) image from FDSS software.

FDSS software option High speed acquisition option U8524-11

Software module and protection key enabling high speed capture.

FDSS software option Interface for external control U8524-13A

Enables FDSS external control interface of FDSS software.

Recommended configuration for semi-automation

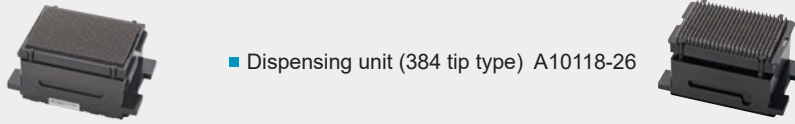
Standard configuration

- FDSS-GX Kinetic Plate Imager C15711-02
 - Fluorescence luminescence sensor unit (1 lamp)
 - Detector
 - Plate loading line
 - Washing stage
 - Plate shaking function
 - Heater unit
 - FDSS software
- Data analysis unit C7903-13
- Automatic tip loader A15623-07



Dispensing units

- Dispensing unit (1536 tip type) A15623-28
- Dispensing unit (384 tip type) A10118-26



Recommended options

- Plate stacker set with lid opener A15623-05
- Barcode reader set for front and rear loaders A15623-50
- Automatic reagent feeder A15623-55
- Deep reservoir A10118-61
- High power ultrasonic tip washer A15623-49
- Dry wipe A15623-56
- FDSS software Additional offline software license U8524-03A
- FDSS software option High speed acquisition option U8524-11

This is an example of a configuration for semi-automation.
Please contact your Hamamatsu representative for further information.

Other components

Dispensing unit	Options
Dispensing unit (96 tip type) A10118-24	Filter set (VSP) A10343-01C
	Filter set (CFP/YFP-FRET) A10343-21B
	Filter set (Fura-2) A10343-61
	FDSS software option Waveform analysis software for cardiomyocyte U8524-12
	FDSS software option Interface for external control U8524-13A
	FDSS software option Export TIFF image option U8524-14

Maintenance and validation service

Maintenance for the hardware and quality check of the dispenser head should be performed periodically to validate your instrument.
The maintenance service and validation service should be done within the first year after installation, and we strongly recommend signing up for a full-service contract that covers the maintenance service and validation service, to certify the instrument's performance. The full-service contract is only offered during the first year after installation. Please contact your Hamamatsu representative for further information.

System appearance



* Stand for Automatic tip loader is not included in the FDSS-GX configuration.

Specifications

Dispensing units	1536 tip type A15623-28	1 µL to 5 µL
	384 tip type A10118-26	1 µL to 30 µL
	96 tip type A10118-24	10 µL to 200 µL
Fluorescence and luminescence detector	Hamamatsu qCMOS sensor dedicated to FDSS	
Number of sampling data points	1 to 50 000 sampling	
Sampling rate	10 Hz (10 data points per second) minimum 120 Hz (120 data points per second) maximum*	
Sampling interval	0.1 s minimum 0.0083 s minimum*	
Excitation light source	Xe lamp Optical filters can be selected	
Plate shaking function	Adjustable in 256 steps Revolutions per minute: 0 to 3000 rpm	
Heater unit	Temperature control part Configurable temperature	
	FDSS-GX Main unit chamber (whole upper half) +40 °C maximum	
Tip loading	Use Automatic tip loader A15623-07	
Number of plate loading lines	Front loader × 1 Rear loader × 1	
Number of plate positions	Assay plate Front loader × 1 Compound plate Rear loader × 1, Compound plate stage (manual exchange) × 3	
Number of tip washer positions	3 units maximum (Total with/without high power ultrasonic function, deep reservoir)	
Number of dispensing unit positions	2 units maximum (same tip type only)	
Compatible dispensing tip	1536 / 384 / 96 black tips dedicated to FDSS, A8687 series Disposable type	
Compatible microplate	Clear bottom black 1536 / 384 / 96 plates SBS format height 10 mm minimum	
Power supply specifications	Single phase AC 100 V to 240 V, 50 Hz / 60 Hz	
Power consumption	C15711-02: approx. 3115 VA C7903-13: approx. 660 W Required supply line: 15 A × 3 lines	
Ambient operating temperature	+15 °C to 30 °C	
Dimensions/ Weight	Main unit	1437 (W) × 850 (D) × 1467 (H) mm / approx. 320 kg
	Data analysis unit	570 (W) × 880 (D) × 943 (H) mm / approx. 82 kg
	Plate stacker set	437 (W) × 580 (D) × 1340 (H) mm / approx. 36 kg
	Washing rack	616 (W) × 924 (D) × 1038 (H) mm / approx. 19 kg
	Circulating water cooler	222 (W) × 386 (D) × 649 (H) mm / approx. 26 kg
	Automatic tip loader	233 (W) × 289 (D) × 447 (H) mm / approx. 21 kg

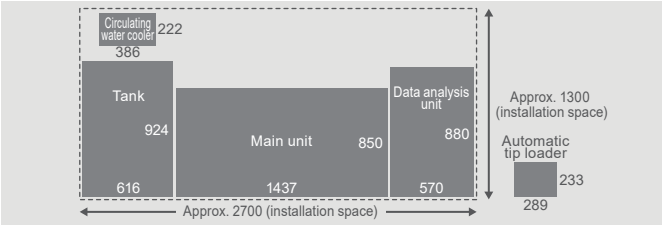
* When using FDSS software option High speed acquisition option U8524-11

	User application					
	Excitation filter 1	Excitation filter 2	Shortpass filter	Dichroic mirror	Emission filter 1	Emission filter 2
Fura-2 (Ca ²⁺)	340 nm	387 nm	SPF 495	UV	520 nm to 560 nm	-
SBFI (Na ⁺)	340 nm	387 nm	SPF 495	UV	520 nm to 560 nm	-
PBFI (K ⁺)	340 nm	387 nm	SPF 495	UV	520 nm to 560 nm	-
MQAE, diH-MEQ (Cl ⁻)	387 nm	-	SPF 495	UV	440 nm to 470 nm	-
DAPI (DNA)	387 nm	-	SPF 495	UV	440 nm to 470 nm	-
Hoechst33258	387 nm	-	SPF 495	UV	440 nm to 470 nm	-
VSP-1	387 nm	-	SPF 400	For VSP	466 nm	560 nm
CFP/YFP	438 nm	-	SPF 450	For C/Y	483 nm	542 nm
HYPHER	425 nm	483 nm	SPF 495	For HYPHER	520 nm to 560 nm	-
Fluo-8 (Ca ²⁺)	472 nm	-	SPF 495	B	520 nm to 560 nm	-
Cal-520 (Ca ²⁺)	472 nm	-	SPF 495	B	520 nm to 560 nm	-
Sodium Green (Na ⁺)	472 nm	-	SPF 495	B	520 nm to 560 nm	-
FluoOR (K ⁺)	472 nm	-	SPF 495	B	520 nm to 560 nm	-
BCECF (pH)	472 nm	-	SPF 495	B	520 nm to 560 nm	-
FluoVio	472 nm	-	SPF 495	B	520 nm to 560 nm	-
GFP	472 nm	-	SPF 495	B	520 nm to 560 nm	-
FITC	472 nm	-	SPF 495	B	520 nm to 560 nm	-
YFP	472 nm	-	SPF 495	For YFP	520 nm to 560 nm	-
JC-1	531 nm	-	SPF 550	For JC-1	593 nm	-
CoroNa Red (Na ⁺)	531 nm	-	SPF 550	For FMP	593 nm	-
FMP	531 nm	-	SPF 550	For FMP	593 nm	-
Cal-590 (Ca ²⁺)	531 nm	-	SPF 550	For FMP	593 nm	-
Rhodamine	531 nm	-	SPF 550	For FMP	593 nm	-
Cal-630 (Ca ²⁺)	605 nm	-	SPF 630	For RED	676 nm	-
Nano-BRET	-	-	-	-	461 nm	647 nm

Standard Options

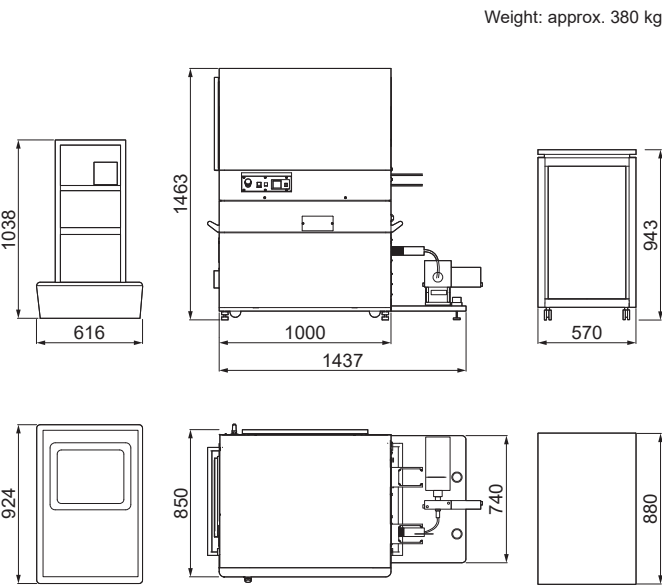
* The wavelength value in the table above may not correspond to the typical excitation/emission wavelength for each application.
Depending on the application, we offer filter sets suitable for the FDSS-GX optical system. Please contact your Hamamatsu representative for further information.

System footprint



* Please secure a separate space to install the automatic tip loader.

Dimensional outlines



Weight: approx. 380 kg

Series lineup

FDSS-µCELL v2

Kinetic Plate Imager
96 / 384 tip type

- Compatible with 96 / 384 microplates
- CO₂ incubator
- EFS pacing system



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